

# PAGE\_31\_FREE\_TOOLS

**URL:** <https://securityblogs.com.au/free-tools/> **Airtable table:** SB · 31 · Free Tools Hub **Page kind:** tool **Sections in spec:** 5

## Business Purpose

Free tools collection: Visibility Checker (form), AI Visibility Checklist (download), Guides tabs.

## Sections (top to bottom)

| # | Section                   | Block type    |
|---|---------------------------|---------------|
| 1 | Hero                      | hero          |
| 2 | Visibility Checker widget | client widget |
| 3 | Checklist download        | rich-text     |
| 4 | Guides tabs               | tabs (client) |
| 5 | CTA band                  | cta-band      |

## Airtable Tables

| Table                        | Use                                                     |
|------------------------------|---------------------------------------------------------|
| SB · 31 · Free Tools Hub     | Per-section spec for /free-tools/ (one row per section) |
| SB · 00 · Pages Index        | Master index entry                                      |
| SB · DOCS · Index (proposed) | Links this MD/PDF to the Airtable page record           |

## PostgreSQL Tables (used by this page)

| Table | Use                                            |
|-------|------------------------------------------------|
| leads | Form submissions land here as new Lead records |
| media | Image attachments (hero, covers, logos)        |

## API Endpoints

| Endpoint                                        | Method           | Cached   | Purpose                  |
|-------------------------------------------------|------------------|----------|--------------------------|
| GET /api/globals/settings                       | GET              | ISR 600s | Brand + contact + footer |
| GET /api/redirects?where[isActive][equals]=true | GET (middleware) | 60s mem  | Redirect rules           |
| POST /api/leads                                 | POST             | dynamic  | Form submission          |

## Request Schemas

### POST /api/leads

```
{
  "name": "string (1-120 chars, required)",
  "email": "string (valid format, required)",
  "phone": "string (optional)",
  "company": "string (optional)",
  "subject": "string (optional)",
  "message": "string (optional, max 2000 chars)",
  "source": "visibility-checker",
  "meta": { "domain": "example.com" },
  "cf-turnstile-response": "string (Cloudflare token, required in production)",
  "company_url": "string (honeypot – must be empty)"
}
```

## Response Schemas

### POST /api/leads

```
// 200 OK
{ "ok": true, "id": "rec..." }

// 400 Bad Request
{ "ok": false, "error": "Email looks invalid" }

// 429 Too Many Requests
{ "ok": false, "error": "Too many submissions. Please wait 15 minutes and try again." }

// 502 Bad Gateway (CMS write rejected)
{ "ok": false, "error": "Could not save your enquiry. Please try again." }
```

## React Components

| Component                            | Purpose                                      |
|--------------------------------------|----------------------------------------------|
| app/free-tools/page.tsx              | Server component — fetches CMS data, renders |
| app/free-tools/VisibilityChecker.tsx | Component used on this page                  |
| app/free-tools/Checklist.tsx         | Component used on this page                  |
| app/free-tools/GuidesTabs.tsx        | Client component — interactive tabs          |
| components/layout/Navbar.tsx         | Component used on this page                  |
| components/layout/Footer.tsx         | Component used on this page                  |
| components/layout/Navbar.tsx         | Shared header                                |
| components/layout/Footer.tsx         | Shared footer                                |

## Node.js Services

| Service / module  | Purpose                               |
|-------------------|---------------------------------------|
| lib/cms.ts        | Typed Payload REST client — all reads |
| middleware.ts     | Edge middleware — CMS redirect rules  |
| lib/submitForm.ts | Browser-side POST helper              |

| Service / module             | Purpose                                                |
|------------------------------|--------------------------------------------------------|
| app/api/leads/route.ts       | Server endpoint — Turnstile + rate-limit + Lead create |
| cms/src/collections/Leads.ts | Lead schema + afterChange email hook                   |
| cms/src/email/transport.ts   | Hostinger SMTP via nodemailer                          |

## Environment Variables

| Var                            | Required?  | Purpose                       |
|--------------------------------|------------|-------------------------------|
| NEXT_PUBLIC_SITE_URL           | Yes        | Canonical site URL            |
| CMS_URL                        | Yes        | Where lib/cms.ts fetches from |
| PAYLOAD_API_KEY                | Yes (prod) | Server-side CMS write auth    |
| NEXT_PUBLIC_TURNSTILE_SITE_KEY | Yes (prod) | Cloudflare widget site key    |
| TURNSTILE_SECRET_KEY           | Yes (prod) | Server-side verify            |
| SENTRY_DSN                     | Yes (prod) | Error tracking                |

## Validation Rules

### /api/leads **validation pipeline**

1. **Honeypot:** company\_url must be empty. If filled, return { ok: true } silently (do NOT create lead).
2. **Shape:** name and email required. email must match /^[^@\\s]+@[^@\\s]+\\. [^@\\s]+\$/.
3. **Rate limit:** Per-IP, 15 submissions per 15 minutes. Returns 429 over limit.
4. **Turnstile:** Server-side siteverify against Cloudflare. Skipped if TURNSTILE\_SECRET\_KEY unset (dev mode).
5. **Source:** Must be one of the SOURCE\_OPTIONS values defined in cms/src/collections/Leads.ts.
6. **Forensic:** IP, user-agent, referer captured automatically.

Source-specific validation runs in the Next.js layer before the CMS write (zod schemas).

## Authentication Rules

- Public POST allowed for browser-side form submission.
- Bot protection: honeypot + Turnstile + per-IP rate limit (15/15min).
- Server-to-server CMS write uses Authorization: users API-Key <PAYLOAD\_API\_KEY>.
- No NextAuth session required from the visitor.

## Deployment Notes

- Server-rendered Next.js page (ISR). NOT compatible with output: 'export'.
- First deploy requires pnpm payload migrate + relevant pnpm seed:\* before booting.
- Caddy reverse-proxies port 3000 (marketing) — see DEPLOYMENT\_GUIDE §6.
- Cloudflare Turnstile site key + secret must be configured in production env.

- Hostinger SMTP credentials must be set in `cms/.env` for assignee notification emails.
- AI score / visibility checker endpoints rely on MCP tools (Ahrefs Brand Radar, DataForSEO) — credentials server-only.

## Testing Procedures

### Local smoke

1. `pnpm dev` (marketing site at `http://localhost:3000`)
2. Navigate to `/free-tools/`
3. Verify hero renders, navigation works, no console errors
4. Fill the form with valid data → expect 200 OK + lead in `/admin/collections/leads`
5. Fill the form with empty email → expect 400
6. Submit 16 times in 15 min from same IP → 16th returns 429
7. Open with JS disabled → confirm graceful no-JS fallback message
4. Verify widget mounts client-side
5. Trigger the widget action → expect cached or computed result

### Production checks

- Lighthouse Performance  $\geq 85$  on landing pages
- Lighthouse Accessibility  $\geq 95$
- Lighthouse SEO = 100
- Sentry: no new error fingerprints after deploy
- Uptime monitor green

## Troubleshooting

| Symptom                        | Likely cause                       | Fix                                                                  |
|--------------------------------|------------------------------------|----------------------------------------------------------------------|
| Page returns 500               | CMS unreachable                    | Check <code>CMS_URL</code> + CMS service status                      |
| Content stale after admin edit | Cache not purged                   | <code>revalidateTag()</code> or wait for revalidate interval         |
| Sitemap missing this page      | CMS Pages record not published     | Set <code>status=published</code>                                    |
| Navbar/footer shows defaults   | Settings global missing            | Run <code>pnpm seed:settings</code>                                  |
| Form submits but no Lead       | <code>PAYLOAD_API_KEY</code> wrong | Regenerate in admin → Users → API Key                                |
| Form rejected with 400 captcha | Turnstile token bad                | Verify production <code>TURNSTILE_SECRET_KEY</code> matches site key |
| Assignee not getting email     | SMTP creds wrong                   | Test Hostinger SMTP from VPS shell                                   |

## Change Impact Analysis

If you change...

- **Copy / hero text:** Low impact. Edit the CMS record (or `app/free-tools/page.tsx` if still hardcoded). Wait for ISR.
- **Sections order:** Medium impact. Re-order modules in admin. Renders instantly on next ISR.
- **Add a new module type:** High impact. Add block to `cms/src/collections/Pages.ts`, add component to `components/modules/blocks/`, add dispatch in `ModuleRenderer.tsx`.
- **Slug: CRITICAL — breaks every URL.** Add a `Redirects` record from old slug → new BEFORE updating. Notify SEO if any external links exist.
- **Authentication scope:** Edit `cms/src/access/*.ts`. Roles changes only by `super_admin`.
- **API endpoint shape:** Breaking change. Bump version, dual-route for 30 days, document migration.

## Rollback Procedures

1. **CMS content rollback:** Use Payload versions UI (`/admin/collections/pages/<id>` → Versions tab). Revert to a prior snapshot.
2. **Code rollback:** `git revert <bad-commit>` on `phase-c-frontend-rewire`, push, redeploy.
3. **Schema migration rollback:** Restore Postgres from nightly backup (see `DEPLOYMENT_GUIDE §7`).
4. **Deploy rollback:** `systemctl stop sb-web && git checkout <prev-commit> && pnpm build && systemctl start sb-web`. <5 min.
5. **DNS-level rollback (emergency only):** Revert A record to the last static-export Hostinger IP (5-minute propagation).

## Forms

- **Visibility Checker** — posts to `/api/leads` (see Request/Response schemas)
- **Checklist email gate** — posts to `/api/leads` (see Request/Response schemas)

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Notes: Visibility Checker posts to `/api/leads` with `source='visibility-checker'` and `meta.domain + meta.score`.

End of `PAGE_31_FREE_TOOLS.md`